Water Supply & Fleet Billing Management System

Project Overview

Summary, modules and workflow diagrams

Prepared for	Shree Karunambika Traders
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1. Summary

Shree Karunambika Traders sells water at a filling point with three dispensing meters. Commercial tankers, each owned by a water supplier such as Sangeetha Water Supply, arrive, fill and leave.

The proposed system records every fill against a continuous meter chain, so that water cannot leave a meter without being accounted for. Each fill becomes an invoice. Each invoice posts automatically to a double-entry ledger. Supplier, vehicle, customer and staff statements are then available in standard accounting format.

The system is built on four principles:

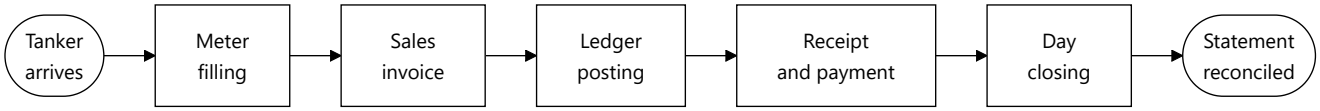
1. **The meter chain is the source of truth.** Each meter's readings form an unbroken sequence. Each vehicle's start is locked to the previous vehicle's end, so any unrecorded water appears as a variance.
2. **The ledger is permanent.** Mistakes are corrected with reversal entries. Balances are never edited.
3. **Manual changes are controlled.** They are available only after 6:00 PM, always need a reason, and permanently mark the record.
4. **One posting, many views.** Supplier, vehicle and customer statements read the same ledger lines, so they cannot disagree.

2. What the system covers

Module	What it does
Dashboard	Daily sales, cash and credit sales, trips, receipts, payments, outstanding amount, cash balance, and live status of Meters 1, 2 and 3, with direct links to invoicing
Vehicle Master	Suppliers, vehicles, unique vehicle codes, rates, drivers, customers and staff
Meter Dispensing	Live record of each fill: In Time, Out Time, Meter ID, start and final readings, and volume
Sales Invoice	Trip invoices and consolidated invoices for a supplier's whole fleet, on cash or credit
Invoice Display	All invoices, receipts and payments for a period, with cash roll-forward
Transactions	Receipt and payment vouchers by Cash, Bank or UPI, with live balance preview
Ledger Master	Expense and other ledgers, such as Vehicle Maintenance, Fuel Expense and Oil Change
Ledgers	Standard statements of account for suppliers, vehicles, customers, staff, cash, bank and expenses
View Day Sales	Day Opening, Daily Transactions and Day Closing, including actual cash count and difference
Reports	Sales, outstanding and ageing, meter reconciliation and override reports, in PDF and Excel
Admin	User accounts, role permissions and a permanent audit log

3. Workflow

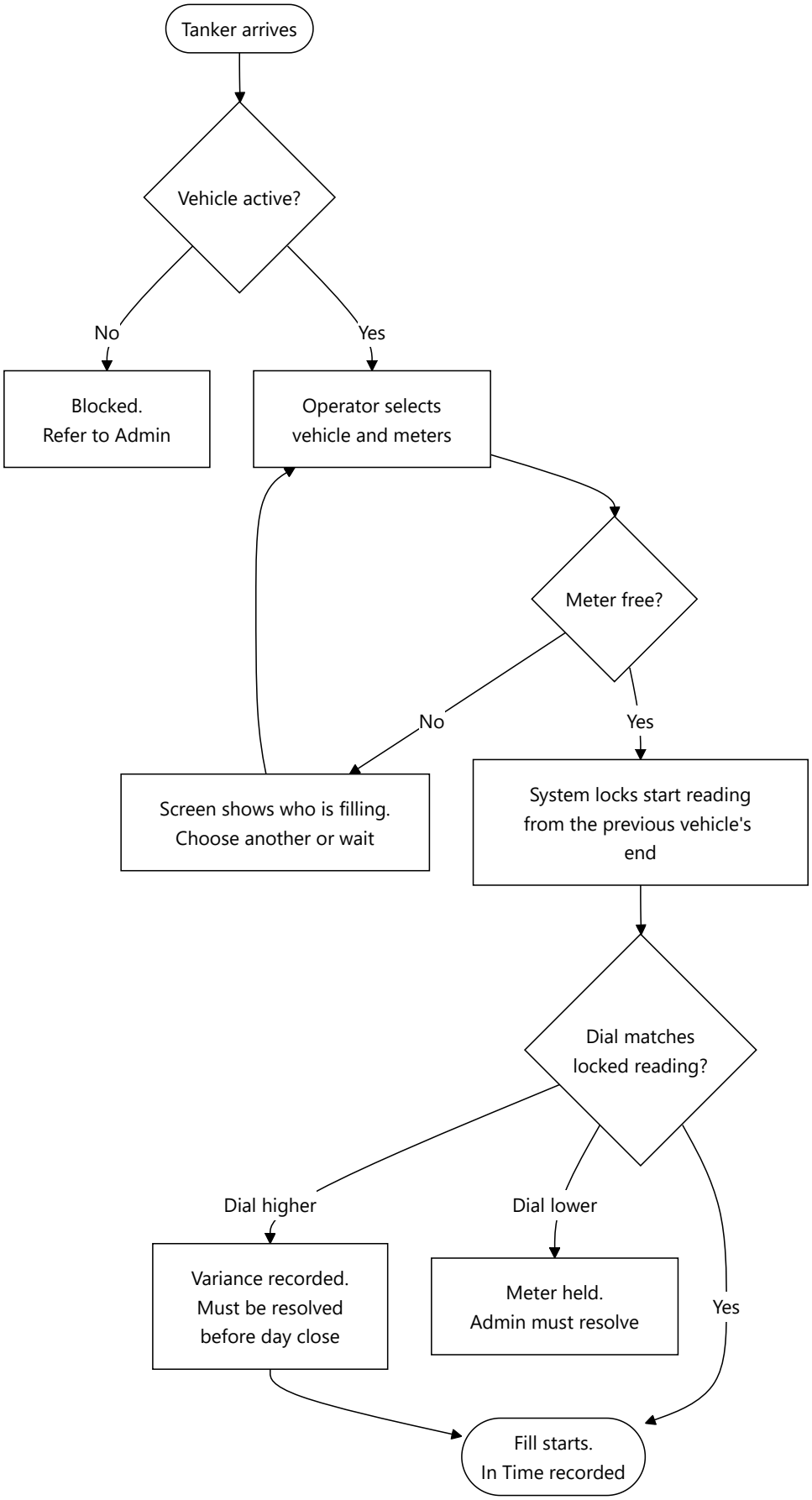
3.1 The whole process at a glance



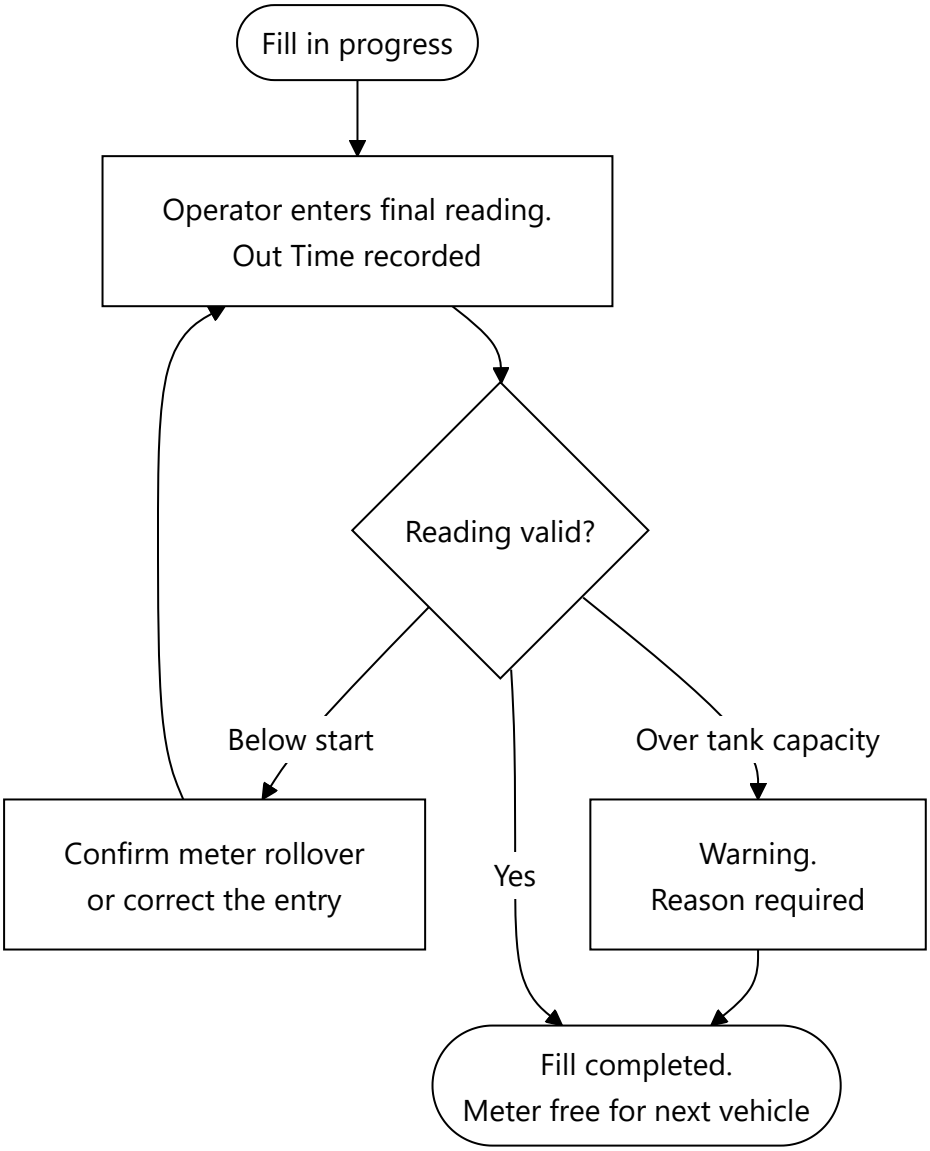
The seven steps below are the same every day. The stages that follow show each step in detail.

Step	What happens	Who
1. Tanker arrives	The operator selects the vehicle. Inactive vehicles are blocked.	Dispatch Operator
2. Meter filling	The start reading locks from the previous vehicle. In Time and Out Time are recorded by the system.	Dispatch Operator
3. Sales invoice	A trip invoice is created, or the trip is held for a consolidated invoice.	Operator or Billing Clerk
4. Ledger posting	The system posts debit and credit entries automatically. No separate data entry is needed.	Automatic
5. Receipt and payment	Money received and paid is entered by Cash, Bank or UPI, and receipts are allocated to invoices.	Billing Clerk or Accountant
6. Day closing	After 6:00 PM, corrections are made, cash is counted and the day is locked.	Accountant
7. Statement reconciled	Supplier and vehicle statements are issued with opening balance, entries and closing balance.	Billing Clerk or Accountant

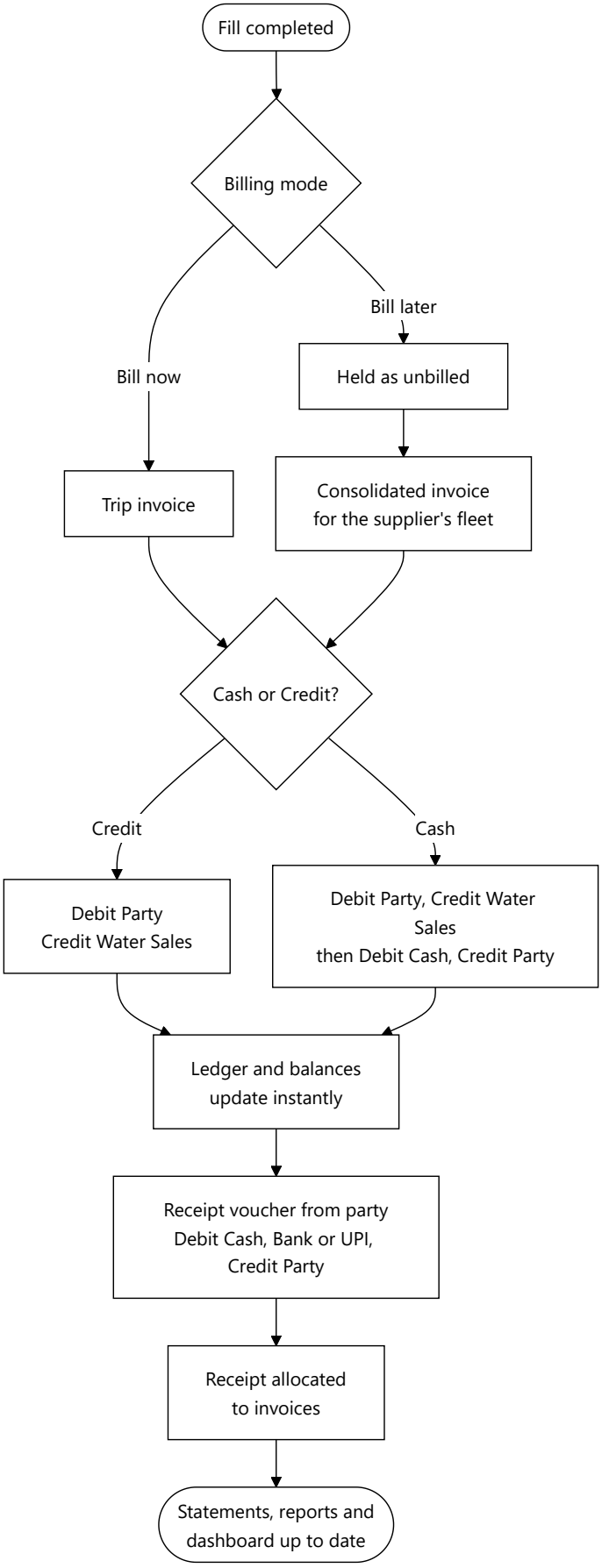
3.2 Stage 1: Tanker arrival and meter start



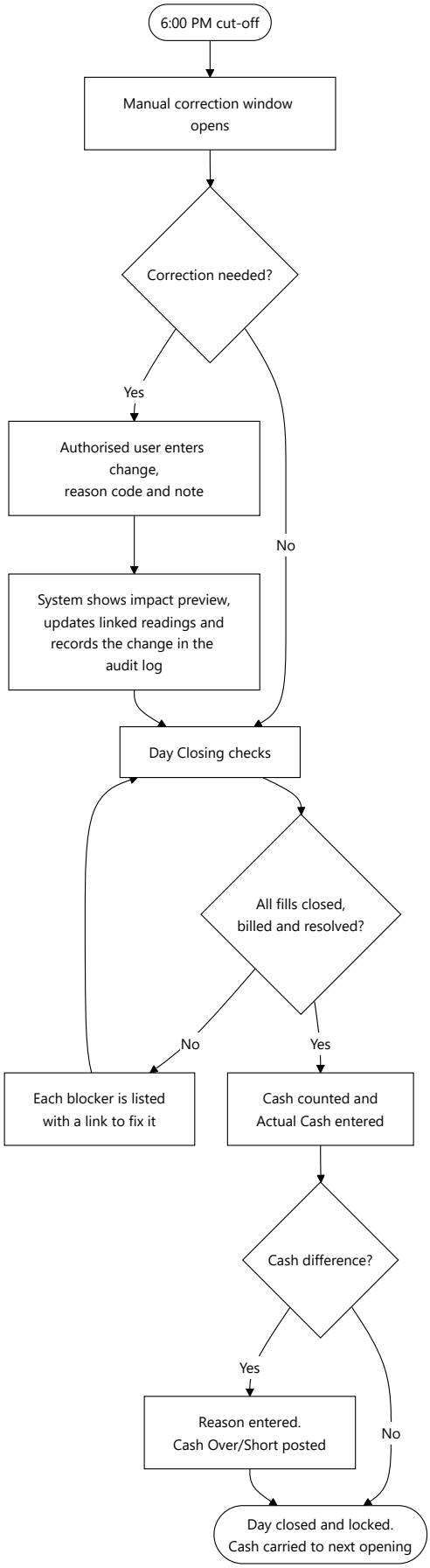
3.3 Stage 2: Filling and completion



3.4 Stage 3: Billing and ledger posting

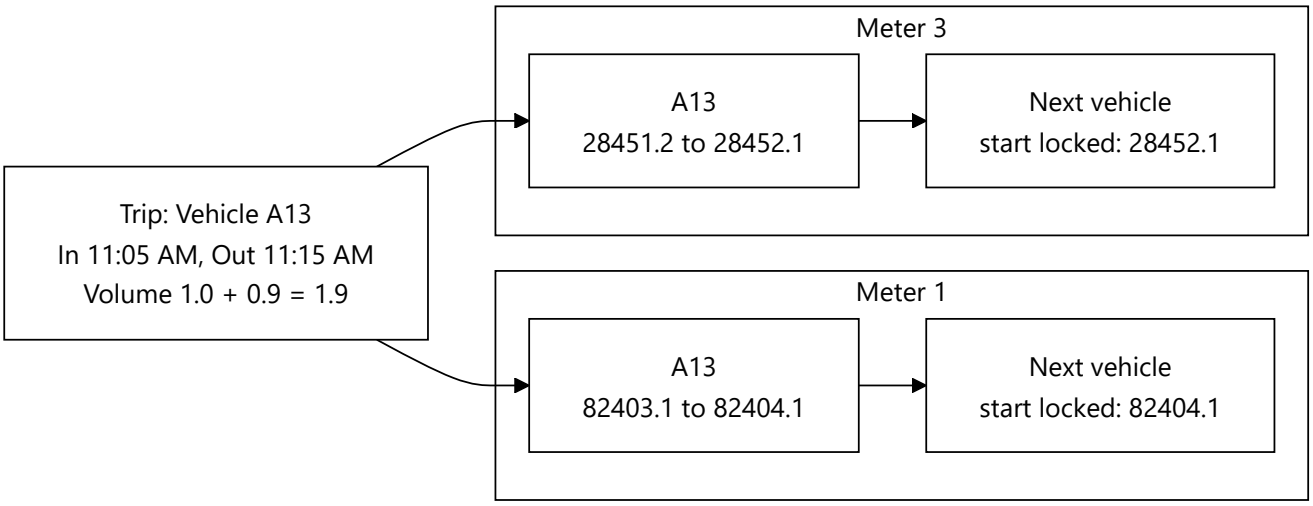


3.5 Stage 4: Evening correction and day closing



3.6 How the meter chain works

Using invoice 7232 from the designs, vehicle A13 draws from Meter 1 and Meter 3. The next vehicle's start on Meter 1 is locked to A13's end.



Each meter has its own chain. Three vehicles can fill at the same time on three meters, and one vehicle can draw from more than one meter. A meter serves only one vehicle at a time.

If the dial disagrees with the chain, the difference is never hidden. It is recorded as a variance, and the day cannot close until that variance is assigned to a vehicle or written off by the Admin.

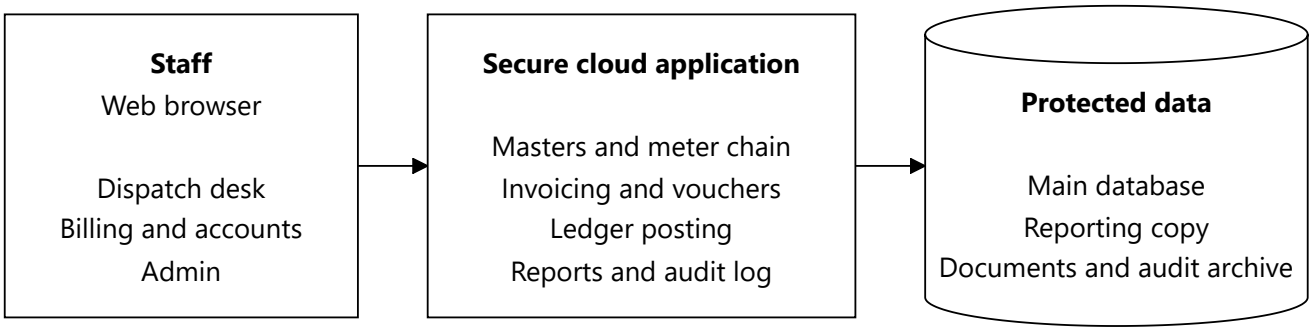
4. Manual correction rules

Rule	Detail
When	From 6:00 PM to 11:59 PM, by server time. Times on staff computers have no effect
Who	Users holding the correction permission. Changes that affect a posted amount need an Admin or Accountant
Required	Reason code and a written note
Effect	The next vehicle's linked reading updates. If an amount changes, the old entry is reversed and a corrected entry posted
Record	Before and after values, user, time and device are kept permanently, and the record shows a MANUAL mark

5. Who uses the system

Role	Main responsibilities
Admin	Users, rates, reopening a closed day, reviewing the audit log
Dispatch Operator	Running fills on the three meters, creating trip invoices
Billing Clerk	Invoices, consolidated billing, supplier statements
Accountant	Vouchers, ledgers, reconciliation, day closing

6. System design



The system runs as one secure web application on a cloud service located in India. Staff use a browser, with nothing to install on each computer. The database is backed up automatically, and audit records are copied daily to write-once storage.

During an internet outage at the site, staff record fills on paper and enter them in the evening correction window with the reason Power or Network Outage. A local server is an option if outages are frequent.

7. Built-in checks

The system checks its own accuracy at each day close and on demand.

Check	What must agree	What it catches
Meter movement	Dial movement and recorded volumes	Unmetered water
Billed volume	Recorded volume and sold, variance and written-off volume	Water not billed
Trial balance	Total debits and total credits	Posting faults
Sub-ledger	Party balances and control accounts	Balance errors
Vehicle split	Vehicle balances and supplier balance	Allocation faults
Running balance	Opening plus debits less credits, and closing	Corrupted balances
Cash book	Opening plus cash in less cash out, and expected cash	Cash errors
Audit chain	Each log entry's link to the one before it	Log tampering

8. Points for confirmation

The system is specified on the working assumptions below. Each one changes how the system behaves, so please confirm or correct them. **Points 1 and 2 are needed before database work begins.**

#	Question	Working assumption	If this is different
1	What does Trip Cost ₹2,500 in Vehicle Master represent? Invoices for those vehicles charge ₹50 to ₹150.	A separate value with no ledger effect. Invoices use a new Rate (₹/Trip) field.	If it is paid to the tanker owner per delivery, each delivery needs a supplier payable entry.
2	Do the meters send readings electronically, or does staff read the dial?	Staff read the dial and the system records the times.	With an electronic feed, hardware connection enters the project scope.
3	What unit is the meter reading? Invoice 7232 shows 1.9 for a 12,000 litre tanker.	Kilolitres, to one decimal, set per meter.	The over-capacity check and volume reports depend on this.
4	Do tanker suppliers buy water from Shree Karunambika Traders?	Yes. Suppliers normally owe the business.	If the business buys from them, ledger direction reverses.
5	Does the trading day run from 6 PM to 6 PM?	Yes. Business day 02/11 runs from 01/11 6:00 PM to 02/11 6:00 PM.	Daily totals and day closing move to a midnight boundary.
6	Does GST apply to these sales?	No tax on invoices, subject to the auditor's confirmation.	Invoices, ledger postings and print layouts need tax fields.
7	How are customers such as Meena Constructions billed?	A Bill To field defaults to the vehicle's supplier and can be changed to a customer.	Delivered water would need a separate delivery order process.